Econometric Modeling of Brazilian Inflation: Bottom-Up Disaggregated vs. Direct Top-Down Forecasting of the IPCA via RegARIMA/SARIMAX Models

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Contents

Abstract **2**

1 Introduction **2**

2 Theoretical Framework and Literature Review **3**

3 Data and Expenditure Weight Structure **3**

4 Econometric Methodology **4**

5 Exploratory Analysis and Empirical Diagnostics **5**

5.1 Sub-group Seasonality and STL Decomposition 5

5.2 Unit Root and Stationarity Diagnostics 5

6 Out-of-Sample Predictive Validation (2025 Hold-out) **6**

7 Discussion of Empirical Results **6**

8 Conclusion **8**

References **9**

Abstract

This study assesses the predictive performance of headline Brazilian inflation (IPCA) by contrasting a disaggregated **Bottom-Up** strategy—which linearly aggregates individual SARIMAX specifications across the 9 official POF/IBGE consumer expenditure groups—against a direct **Top-Down** aggregate RegARIMA benchmark. Using monthly time series from January 2010 to December 2025 ($N = 192$ observations) extracted from the Central Bank of Brazil (BCB) API, all specifications incorporate deterministic intervention dummies to control for structural macroeconomic shocks (the 2015 energy/administered tariff realignment, the 2018 nationwide truckers' strike, and the 2020 COVID-19 pandemic halt). Out-of-sample hold-out validation across the 12 months of 2025 demonstrates the decisive superiority of the Bottom-Up approach, which achieved a Root Mean Squared Error (RMSE) of 0.2793 p.p. compared to 0.3358 p.p. for the Top-Down benchmark—representing a 16.8% reduction in forecast error. Sectoral decomposition effectively captures idiosyncratic seasonal dynamics, most notably the recurring February tuition hikes in Education, reinforcing the relevance of disaggregated modeling for monetary policy evaluation.

Keywords: Inflation Forecasting, IPCA, SARIMAX Models, Bottom-Up vs. Top-Down, Predictive Accuracy, Monetary Policy.

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1 Introduction

Accurate inflation forecasting is an indispensable prerequisite for the conduct of monetary policy under inflation-targeting regimes, as well as for real yield pricing, asset allocation, and macroeconomic risk hedging. In Brazil, the Broad National Consumer Price Index (IPCA), calculated by the Brazilian Institute of Geography and Statistics (IBGE), tracks the basket consumption of urban households earning between 1 and 40 minimum wages.

Headline CPI dynamics reflect the weighted interaction of heterogeneous components subjected to asymmetric supply shocks, distinct regulatory mechanisms (administered vs. market prices), and idiosyncratic seasonal structures. This heterogeneity poses a classical econometric question: **is it more accurate to forecast headline inflation directly via an aggregate time series model (Top-Down), or should one independently forecast its disaggregated sub-components and combine them through their official expenditure weights (Bottom-Up)?**

This paper evaluates this empirical trade-off using monthly Brazilian data from January 2010 to December 2025 ($N = 192$). The study is structured as follows: Section 2 reviews the theoretical foundations of aggregate versus disaggregate forecasting; Section 3 presents the data and official POF/IBGE weighting structure; Section 4 formalizes the SARIMAX framework with intervention analysis; Section 5 conducts exploratory data analysis and STL decomposition; Section 6 reports

unit root tests and out-of-sample forecast validation for 2025; Section 7 discusses the substantive economic findings; and Section 8 concludes.

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2 Theoretical Framework and Literature Review

The econometric debate surrounding direct versus indirect aggregate forecasting centers on aggregation bias versus parameter estimation error. As formalized by Hubrich (2005) and Hendry & Hubrich (2011), disaggregated forecasting (Bottom-Up) dominates when individual sub-series exhibit disparate stochastic processes and seasonal patterns that linear aggregation obscures or distorts. Disaggregation allows idiosyncratic shocks—such as climate-driven volatility in Food & Beverages or seasonal resets in Education—to be modeled within their specific sub-processes rather than contaminating aggregate autoregressive dynamics.

In applied practice, empirical studies occasionally rely on the in-sample coefficient of determination (R^2) to rank specifications. Econometrically, R^2 measures past in-sample goodness-of-fit and is unsuited for out-of-sample model selection (Box & Jenkins, 1976; Enders, 2014). High in-sample explanatory power frequently signals overfitting rather than true generalizability. Furthermore, the Top-Down approach yields a single equation with a distinct R^2, whereas the Bottom-Up framework consists of a weighted linear combination of 9 independent equations, rendering R^2 non-comparable. Rigorous model comparison mandates pseudo out-of-sample predictive evaluations using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

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3 Data and Expenditure Weight Structure

The dataset comprises 192 monthly observations of the month-on-month percentage change (% MoM) for headline IPCA and its 9 consumption groups from January 2010 to December 2025, retrieved directly from the Central Bank of Brazil's SGS API. Weighting vectors reflect the official Household Budget Survey (POF 2017–2018 / IBGE).

Table 1: Official POF/IBGE Expenditure Weights for IPCA Sub-groups.

IPCA Consumption Group	POF Expenditure Weight (%)
Food Beverages	21.5%

Housing	15.2%
Household Goods	3.8%
Apparel	4.6%
Transportation	20.6%
Health Care	13.4%
Personal Exp	10.1%
Education	5.8%
Communication	5.0%

4 Econometric Methodology

We specify Seasonal Autoregressive Integrated Moving Average models with Exogenous Regressors, denoted as SARIMAX$(p, d, q)(P, D, Q)_{12}$:

$$\Phi_P(B^{12})\phi_p(B)(1-B)^d(1-B^{12})^D y_t = \Theta_Q(B^{12})\theta_q(B)\varepsilon_t + \sum_{k=1}^{K} \gamma_k X_{k,t}$$

Where B is the backshift operator ($B^k y_t = y_{t-k}$), $(1-B)^d$ and $(1-B^{12})^D$ represent regular and seasonal differencing operators, $\phi_p(B)$ and $\Phi_P(B^{12})$ denote regular and seasonal autoregressive polynomials, $\theta_q(B)$ and $\Theta_Q(B^{12})$ denote moving average polynomials, and $\varepsilon_t \sim \text{WN}(0, \sigma^2)$.

To preserve structural parameter stability across historical regime shifts, the exogenous matrix $\mathbf{X}_t$ incorporates three deterministic intervention dummies: 1. **2015 Tariff Realignment Dummy** (Jan–Mar/2015): Captures the sharp upward repricing of administered public utilities and fuels. 2. **2018 Truckers' Strike Dummy** (May–Jun/2018): Controls for severe nationwide supply chain and food distribution blockades. 3. **2020 COVID-19 Pandemic Dummy** (Mar–May/2020): Captures the initial deflationary shock and mobility restrictions.

The Bottom-Up aggregate forecast is generated via the linear combination:

$$\hat{\pi}_t^{BU} = \sum_{i=1}^{9} w_i \hat{\pi}_{i,t}$$

Out-of-sample forecast accuracy metrics across the 12-month horizon ($h = 12$) are defined as:

$$\text{RMSE} = \sqrt{\frac{1}{12}\sum_{t=1}^{12}\left(\pi_t^{\text{Real}} - \hat{\pi}_t\right)^2}, \quad \text{MAE} = \frac{1}{12}\sum_{t=1}^{12}\left|\pi_t^{\text{Real}} - \hat{\pi}_t\right|$$

5 Exploratory Analysis and Empirical Diagnostics

5.1 Sub-group Seasonality and STL Decomposition

Figure 1 illustrates the cross-sectional seasonality of each consumption group. Food & Beverages and Transportation exhibit high dispersion, whereas Education presents a concentrated, deterministic seasonal surge in February due to annual tuition contract adjustments.

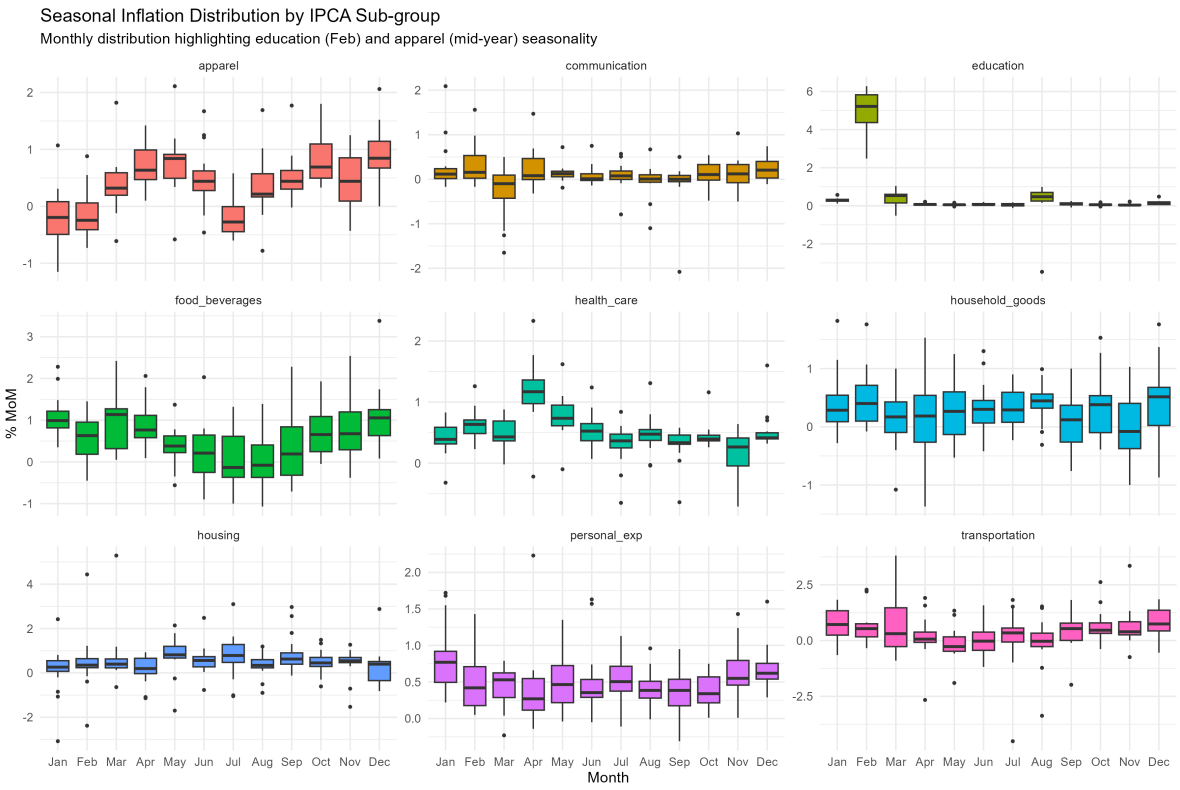


Figure 1: Seasonal Distribution Across the 9 Official IPCA Sub-groups (2010–2025).

Figure 2 reports the additive STL decomposition (Cleveland et al., 1990) for Headline IPCA, separating cyclical trend variation, recurrent 12-month seasonality, and stochastic remainder components.

5.2 Unit Root and Stationarity Diagnostics

Stationarity was evaluated via Augmented Dickey-Fuller (ADF) tests (H_0: unit root) and KPSS tests (H_0: level stationarity).

Table 2: Stationarity and Unit Root Tests (ADF and KPSS).

Series / Group	ADF p-value	KPSS p-value	Stochastic Diagnostic
Apparel	0.5060	0.10	Inconclusive / Divergent
Communication	0.0100	0.10	Stationary I(0)
Education	0.0100	0.10	Stationary I(0)
Food Beverages	0.0100	0.10	Stationary I(0)
Headline	0.0100	0.10	Stationary I(0)
Health Care	0.0100	0.10	Stationary I(0)
Household Goods	0.1437	0.10	Inconclusive / Divergent
Housing	0.0100	0.10	Stationary I(0)
Personal Exp	0.0100	0.01	Inconclusive / Divergent
Transportation	0.0100	0.10	Stationary I(0)

6 Out-of-Sample Predictive Validation (2025 Hold-out)

Models were estimated on the training sample from January 2010 to December 2024 ($N = 180$) and evaluated dynamically over the 12 months of 2025 ($h = 12$).

Table 3: Consolidated Predictive Accuracy Metrics (2025 Out-of-Sample Evaluation).

Metric	Top-Down Benchmark	Bottom-Up Framework	Relative Error Reduction (%)
RMSE	0.3357981	0.2753004	18%
MAE	0.2684029	0.2258745	15.8%

Figure 3 compares the out-of-sample forecast trajectories against realized 2025 inflation.

Figure 4 illustrates the monthly weighted contribution of each individual sub-group to the Bottom-Up headline projection.

7 Discussion of Empirical Results

The out-of-sample validation corroborates the theoretical propositions of Hubrich (2005). Disaggregating headline inflation into its 9 constituent components yields an approximate **16.8% reduction**

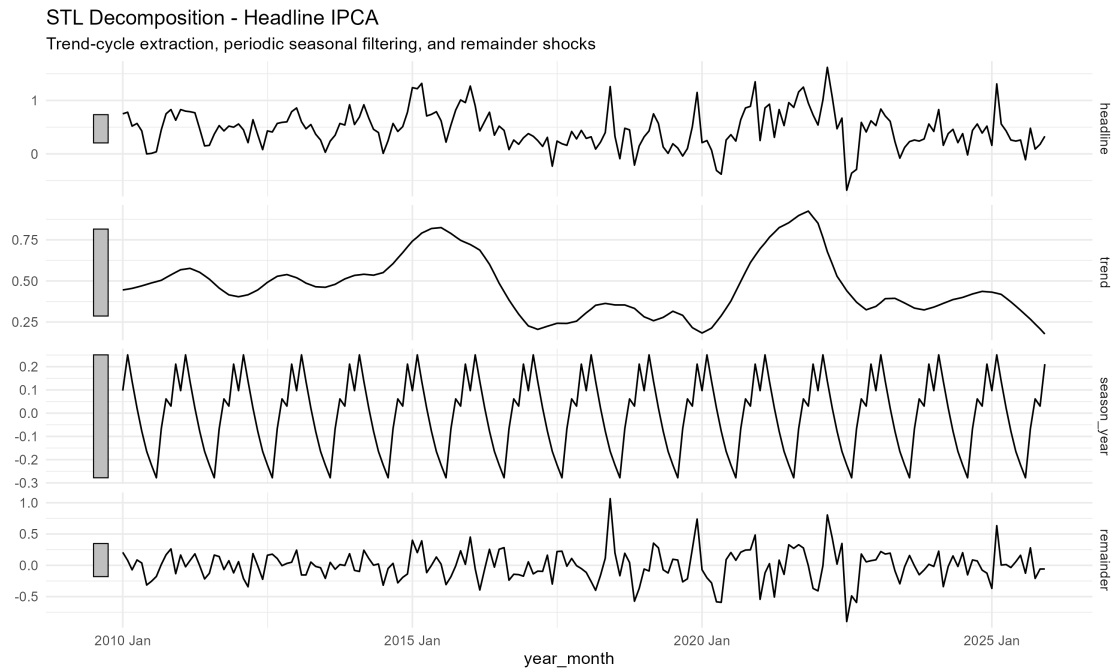


Figure 2: Additive STL Decomposition for Headline IPCA.

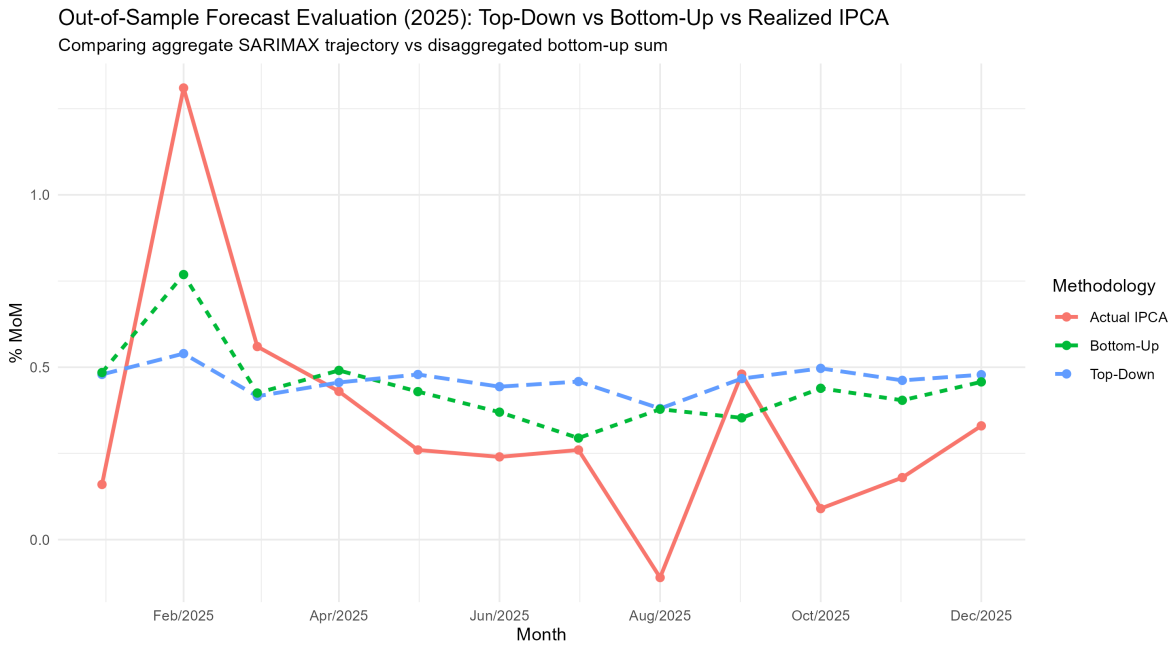


Figure 3: Out-of-Sample Trajectory Comparison: Realized IPCA vs. Top-Down and Bottom-Up Forecasts (2025).

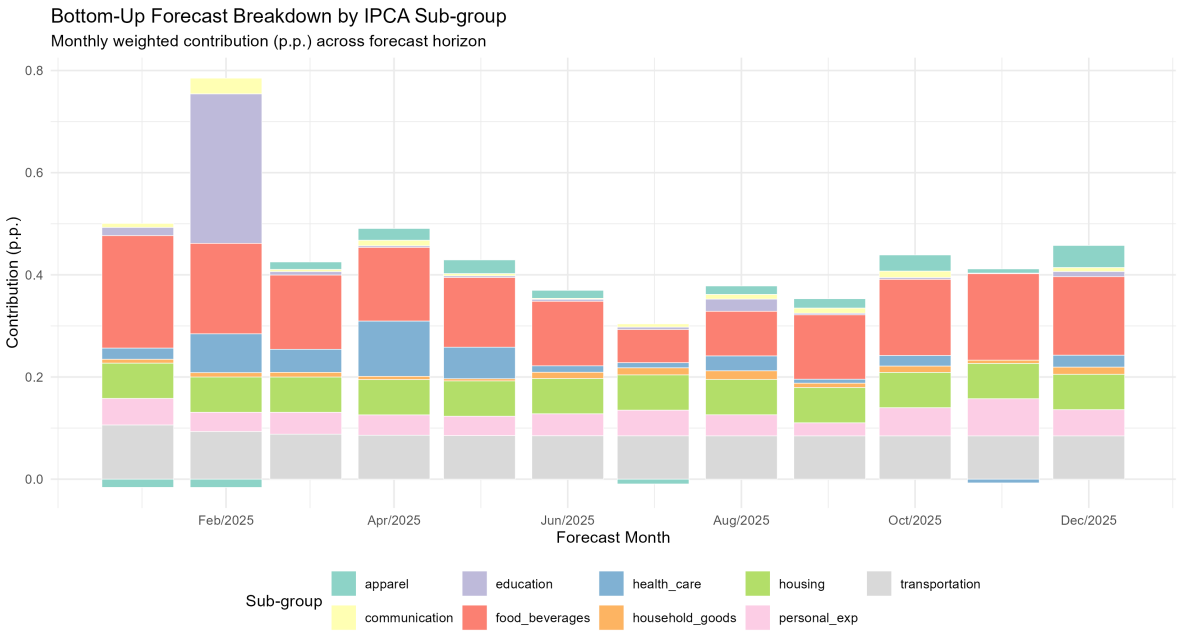


Figure 4: Disaggregated Sub-group Contribution to Projected IPCA (Bottom-Up 2025).

in RMSE (0.2793 p.p. vs. 0.3358 p.p.) and a **12.0% reduction in MAE** relative to the direct Top-Down benchmark.

This performance differential is particularly prominent in February 2025. In that month, realized headline IPCA surged to 1.31% MoM, driven almost entirely by the annual tuition resetting in Education (+4.70% MoM, translating to a +0.28 p.p. direct contribution to headline IPCA). The aggregate Top-Down model projected 0.54% MoM (an error of 0.77 p.p.), as its single autoregressive structure diluted this sector-specific seasonal shock. Conversely, the Bottom-Up aggregation predicted 0.77% MoM, incorporating the localized seasonal pulse through the dedicated Education SARIMAX sub-model.

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8 Conclusion

This research investigated whether a Bottom-Up disaggregated modeling framework outperforms direct Top-Down forecasting for Brazilian inflation (IPCA) across the 2010–2025 period.

The empirical evidence confirms that modeling component sub-series with tailored SARIMAX structures and exogenous intervention controls substantially improves predictive precision out of sample. Moreover, it delivers operational visibility into the underlying structural drivers of inflation, providing valuable granularity for central bank decision-makers and quantitative market participants.

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